

CS2601 Linear and Convex Optimization

Homework 5

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1. $\mathcal{L}(\mathbf{x}, \lambda) = x_1 x_2 + \lambda(x_1^2 + 4x_2^2 - 2)$

The KKT conditions are

$$\begin{cases} \partial_{x_1} \mathcal{L} = x_2 + 2\lambda x_1 = 0 \\ \partial_{x_2} \mathcal{L} = x_1 + 8\lambda x_2 = 0 \\ x_1^2 + 4x_2^2 - 2 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -1 \\ x_2 = \frac{1}{2} \\ \lambda = \frac{1}{4} \end{cases}$$

$$\min f(x_1, x_2) = -\frac{1}{2}$$

2. $\mathcal{L}(\mathbf{x}, \mu_1, \mu_2) = x_1^2 + x_2^2 + \mu_1((x_1 + 1)^2 + (x_2 - 1)^2 - 1) + \mu_2(x_1^2 + (x_2 - 1)^2 - 1)$

The KKT conditions are

$$\begin{cases} \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) + 2\mu_2 x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 - 1) = 0 \\ \mu_1((x_1 + 1)^2 + (x_2 - 1)^2 - 1) = 0 \\ \mu_2(x_1^2 + (x_2 - 1)^2 - 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ x_1^2 + (x_2 - 1)^2 - 1 \leq 0 \end{cases}$$

Case I. Both g_1 and g_2 are inactive, so $\mu_1 = \mu_2 = 0$

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

But

$$(x_1 + 1)^2 + (x_2 - 1)^2 - 1 = 1 > 0, x_1^2 + (x_2 - 1)^2 - 1 = 0$$

violating $g_1 \leq 0$

Case II. g_2 is active, but g_1 is inactive, so $\mu_1 = 0$

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_2 x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_2(x_2 - 1) = 0 \\ x_1^2 + (x_2 - 1)^2 - 1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 2 \\ \mu_2 = -2 \end{cases}$$

violating $\mu_2 \geq 0$

Case III. g_1 is active, but g_2 is inactive, so $\mu_2 = 0$

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 1 = 0 \\ \mu_1 \geq 0 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1}{\sqrt{2}} - 1 \\ x_2 = -\frac{1}{\sqrt{2}} + 1 \\ \mu_1 = \sqrt{2} - 1 \end{cases}$$

And

$$(x_1 + 1)^2 + (x_2 - 1)^2 - 1 = 0, x_1^2 + (x_2 - 1)^2 - 1 = 1 - \sqrt{2} < 0$$

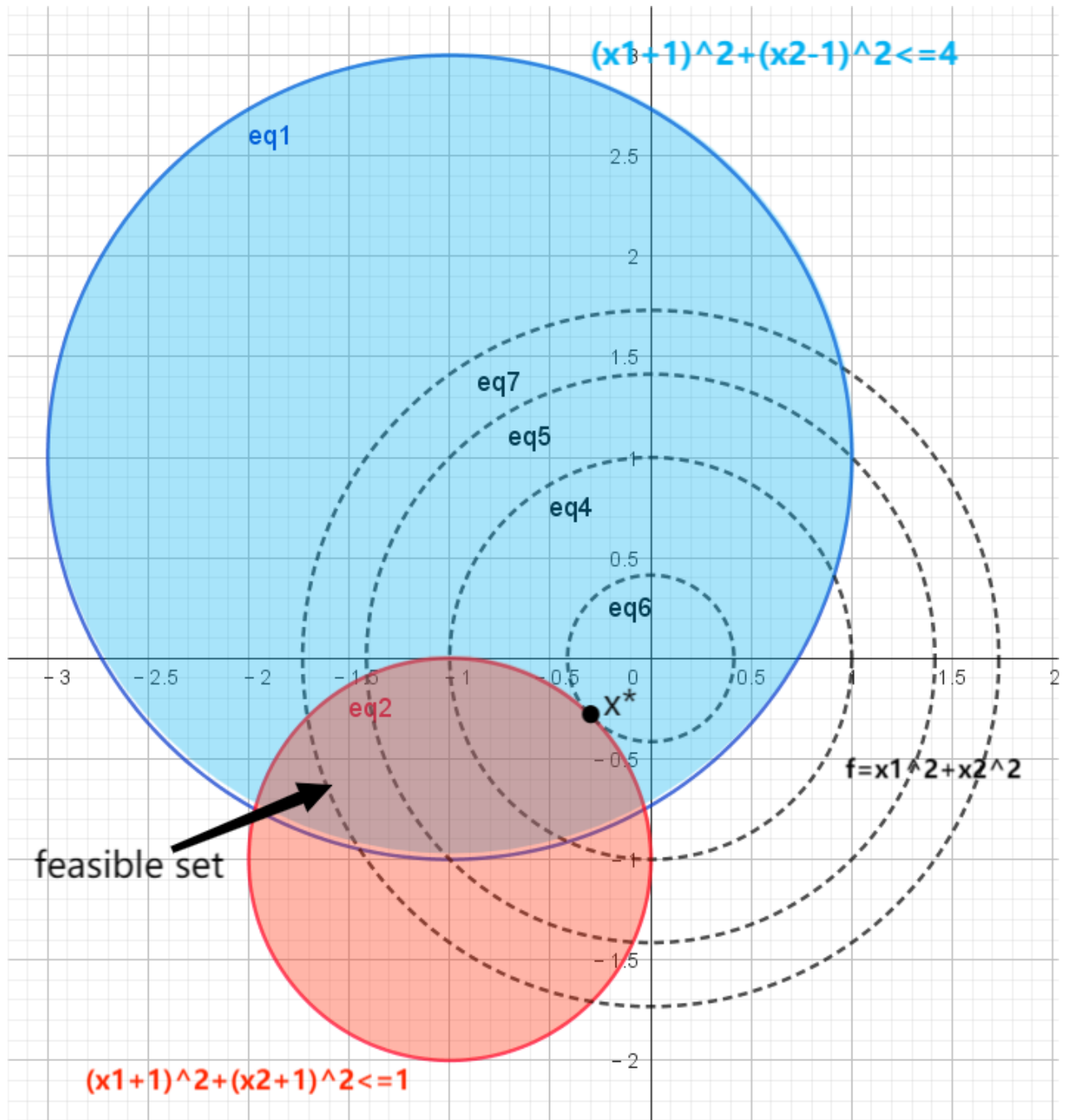
Case IV. Both g_1 and g_2 are active

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) + 2\mu_2 x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 - 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 1 = 0 \\ x_1^2 + (x_2 - 1)^2 - 1 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -\frac{1}{2} \\ x_2 = 1 + \frac{\sqrt{3}}{2} \\ \mu_1 = -\frac{\sqrt{3}}{3} (\text{violating } \mu_1 \geq 0) \\ \mu_2 = -\frac{\sqrt{3}}{3} - 1 (\text{violating } \mu_2 \geq 0) \end{cases} \quad \text{or} \quad \begin{cases} x_1 = -\frac{1}{2} \\ x_2 = 1 - \frac{\sqrt{3}}{2} \\ \mu_1 = \frac{\sqrt{3}}{3} \\ \mu_2 = \frac{\sqrt{3}}{3} - 1 (\text{violating } \mu_2 \geq 0) \end{cases}$$

Therefore $\mathbf{x}^* = (\frac{1}{\sqrt{2}} - 1, -\frac{1}{\sqrt{2}} + 1)$ is the global minimum

3. (a.)



The optimal point $\mathbf{x}^* = (-1 + \frac{\sqrt{2}}{2}, -1 + \frac{\sqrt{2}}{2})$, and the optimal value $f^* = (\sqrt{2} - 1)^2$

(b.) Let $g_1(x_1, x_2) = (x_1 + 1)^2 + (x_2 - 1)^2 - 4$, $g_2(x_1, x_2) = (x_1 + 1)^2 + (x_2 + 1)^2 - 1$

$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = x_1^2 + x_2^2 + \mu_1((x_1 + 1)^2 + (x_2 - 1)^2 - 4) + \mu_2((x_1 + 1)^2 + (x_2 + 1)^2 - 1)$

The KKT conditions are

$$\begin{cases} \mu_1 \geq 0 \\ \mu_2 \geq 0 \\ \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) + 2\mu_2(x_1 + 1) = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 + 1) = 0 \\ \mu_1((x_1 + 1)^2 + (x_2 - 1)^2 - 4) = 0 \\ \mu_2((x_1 + 1)^2 + (x_2 + 1)^2 - 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 4 \leq 0 \\ (x_1 + 1)^2 + (x_2 + 1)^2 - 1 \leq 0 \end{cases}$$

Case I. Both g_1 and g_2 are inactive, so $\mu_1 = \mu_2 = 0$

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

But

$$(x_1 + 1)^2 + (x_2 - 1)^2 - 4 = -2 < 0, (x_1 + 1)^2 + (x_2 + 1)^2 - 1 = 1 > 0$$

violating $g_2 \leq 0$

Case II. g_2 is active, but g_1 is inactive, so $\mu_1 = 0$

$$\begin{cases} \mu_2 \geq 0 \\ \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_2(x_1 + 1) = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_2(x_2 + 1) = 0 \\ (x_1 + 1)^2 + (x_2 + 1)^2 - 1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{\sqrt{2}}{2} - 1 \\ x_2 = \frac{\sqrt{2}}{2} - 1 \\ \mu_2 = \sqrt{2} - 1 \end{cases}$$

And

$$(x_1 + 1)^2 + (x_2 - 1)^2 - 4 = 1 - 2\sqrt{2} < 0, (x_1 + 1)^2 + (x_2 + 1)^2 - 1 = 0$$

Case III. g_1 is active, but g_2 is inactive, so $\mu_2 = 0$

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = \sqrt{2} - 1 \\ x_2 = 1 - \sqrt{2} \\ \mu_1 = -\frac{\sqrt{2}-1}{\sqrt{2}} < 0 \end{cases} \quad or \quad \begin{cases} x_1 = -\sqrt{2} - 1 \\ x_2 = 1 + \sqrt{2} \\ \mu_1 = \frac{-\sqrt{2}-1}{\sqrt{2}} < 0 \end{cases}$$

Violating $\mu_1 \geq 0$

Case IV. Both g_1 and g_2 are active

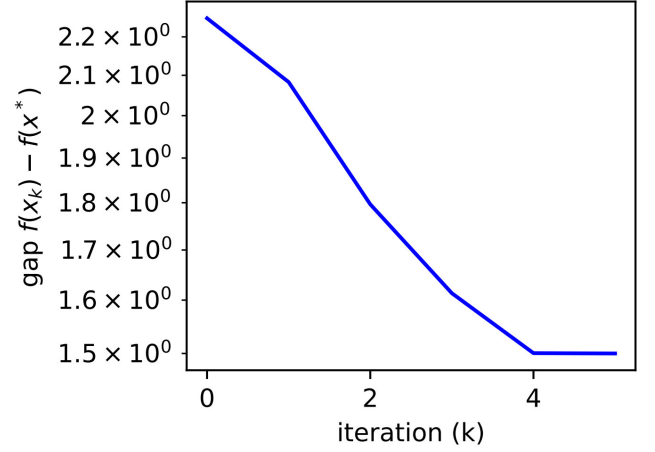
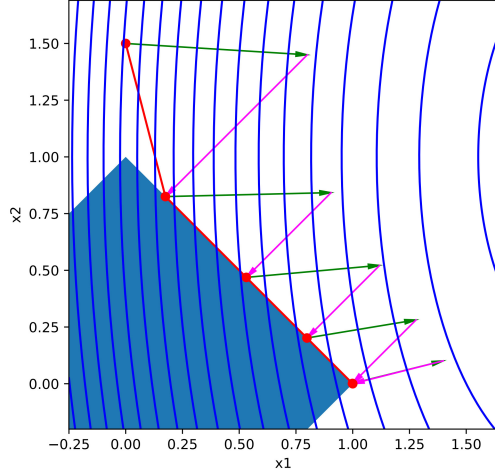
$$\begin{cases} \partial_{x_1} \mathcal{L} = 2x_1 + 2\mu_1(x_1 + 1) + 2\mu_2(x_1 + 1) = 0 \\ \partial_{x_2} \mathcal{L} = 2x_2 + 2\mu_1(x_2 - 1) + 2\mu_2(x_2 + 1) = 0 \\ (x_1 + 1)^2 + (x_2 - 1)^2 - 4 = 0 \\ (x_1 + 1)^2 + (x_2 + 1)^2 - 1 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -1 + \frac{\sqrt{15}}{4} \\ x_2 = -\frac{3}{4} \\ \mu_1 = \frac{1}{2\sqrt{15}} - \frac{1}{2} (violating \quad \mu_1 \geq 0) \\ \mu_2 = \frac{7}{2\sqrt{15}} - \frac{1}{2} \end{cases} \quad or \quad \begin{cases} x_1 = -1 - \frac{\sqrt{15}}{4} \\ x_2 = 1 - \frac{\sqrt{3}}{2} \\ \mu_1 = -\frac{1}{2\sqrt{15}} - \frac{1}{2} (violating \quad \mu_1 \geq 0) \\ \mu_2 = -\frac{7}{2\sqrt{15}} - \frac{1}{2} (violating \quad \mu_2 \geq 0) \end{cases}$$

Therefore there exist Lagrange multipliers that satisfy the KKT conditions, $\mathbf{x}^* = (-1 + \frac{\sqrt{2}}{2}, -1 + \frac{\sqrt{2}}{2})$, $\mu_1 = 0, \mu_2 = \sqrt{2} - 1$
 $\therefore \nabla g_1(\mathbf{x}^*) = (\sqrt{2}, \sqrt{2} - 4)^T, \nabla g_2(\mathbf{x}^*) = (\sqrt{2}, \sqrt{2})^T$.
 $\therefore \nabla g_1(\mathbf{x}^*), \nabla g_2(\mathbf{x}^*)$ are linearly independent.
 $\therefore \mathbf{x}^*$ is regular

4. The solution is as follows:

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number of iterations: 5
solution: [1. 0.]
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5. (a.) $\mathcal{L}(\mathbf{x}, \lambda) = e^{2x_1} + e^{x_2} + e^{x_3} + \lambda(x_1 + x_2 + x_3 - 1)$
 By Lagrange condition,

$$\begin{cases} \partial_{x_1} \mathcal{L} = 2e^{2x_1} + \lambda = 0 \\ \partial_{x_2} \mathcal{L} = e^{x_2} + \lambda = 0 \\ \partial_{x_3} \mathcal{L} = e^{x_3} + \lambda = 0 \\ \partial_{\lambda} \mathcal{L} = x_1 + x_2 + x_3 - 1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{1-2\ln 2}{5} \\ x_2 = \frac{2+\ln 2}{5} \\ x_3 = \frac{2+\ln 2}{5} \\ \lambda = -(2e^2)^{\frac{1}{5}} \end{cases}$$

Therefore $\mathbf{x}^* = (\frac{1-2\ln 2}{5}, \frac{2+\ln 2}{5}, \frac{2+\ln 2}{5})$, $\lambda^* = -(2e^2)^{\frac{1}{5}}$, $f^* = -3(2e^2)^{\frac{1}{5}}$

(b.) Let $\mathbf{A} = [1, 1, 1]$.

For feasible $\mathbf{x}$, the Newton direction is the solution to the approximation

$$\min_{\mathbf{v}} \hat{f}(\mathbf{x} + \mathbf{v}) = f(\mathbf{x}) + (\mathbf{x})^T \mathbf{v} + \frac{1}{2} \mathbf{v}^T \nabla^2 f(\mathbf{x}) \mathbf{v}$$

$$s.t. \mathbf{A}(\mathbf{x} + \mathbf{v}) = 1$$

The KKT system is

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}) & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}) \\ \mathbf{0} \end{bmatrix}$$

$$\Rightarrow \begin{cases} v_1 = \frac{2(e^{-2x_1} - e^{-x_2} - e^{-x_3})}{e^{-2x_1} + 4e^{-x_2} + 4e^{-x_3}} \\ v_2 = \frac{-e^{-2x_1} + 6e^{-x_2} - 4e^{-x_3}}{e^{-2x_1} + 4e^{-x_2} + 4e^{-x_3}} \\ v_3 = \frac{-e^{-2x_1} - 4e^{-x_2} + 6e^{-x_3}}{e^{-2x_1} + 4e^{-x_2} + 4e^{-x_3}} \end{cases}$$

(c.)

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number of iterations: 15
solution: [[-0.07725887]
 [ 0.53862944]
 [ 0.53862944]]
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